Autonomous Vehicle Project



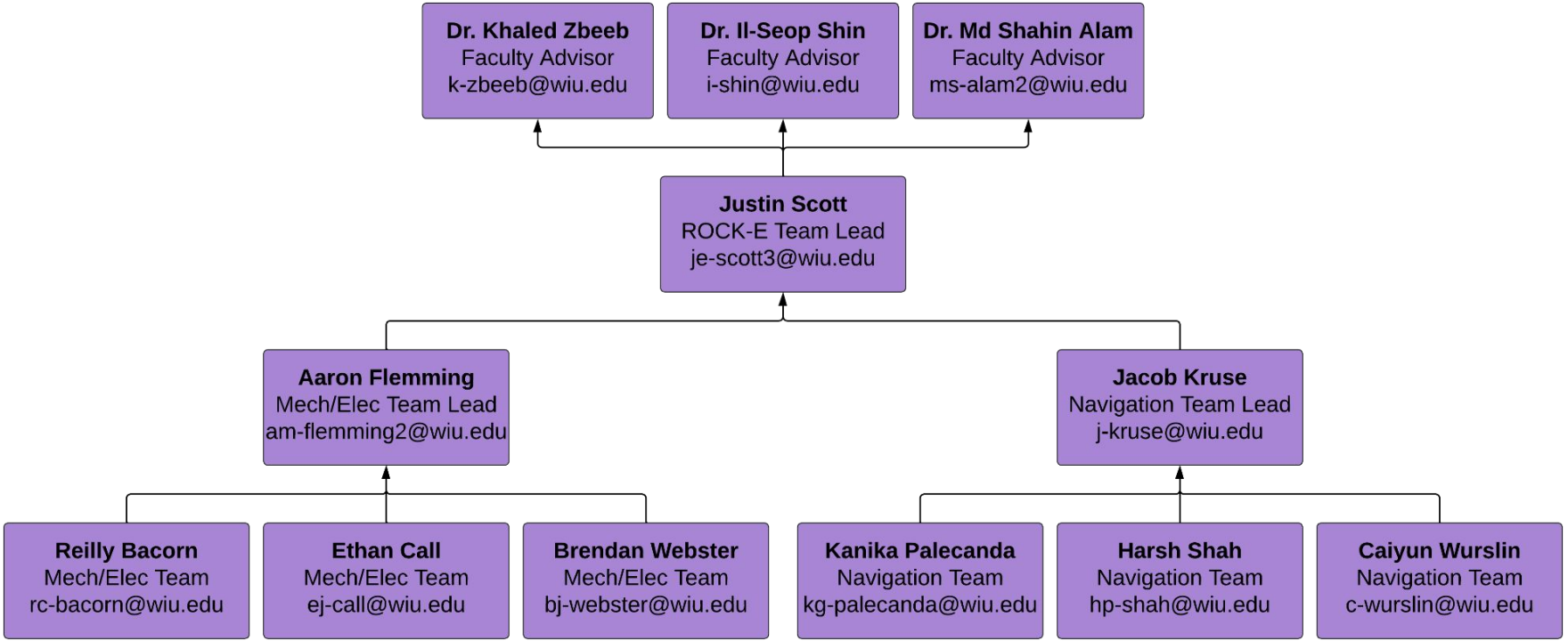
Jacob Kruse · May 28, 2024

Rock-E Team Members

Harsh Shah	Mechanical Engineering
Brendan Webster	Mechanical Engineering
Reilly Bacorn	Mechanical Engineering / Engineering Physics
Kanika Palecanda	Mechanical Engineering
Justin Scott	Mechanical Engineering
Caiyun Wurslin	Electrical Engineering
Ethan Call	Mechanical Engineering
Jacob Kruse	Electrical Engineering
Aaron Flemming	Mechanical Engineering



Team Breakdown



What did we design?

- Vehicle to Compete in 2024 IGVC
- Showcase Vehicle for WIU SET



Why an Autonomous Vehicle?

- Fully Adopted by 2030
- Skills Needed by Industry

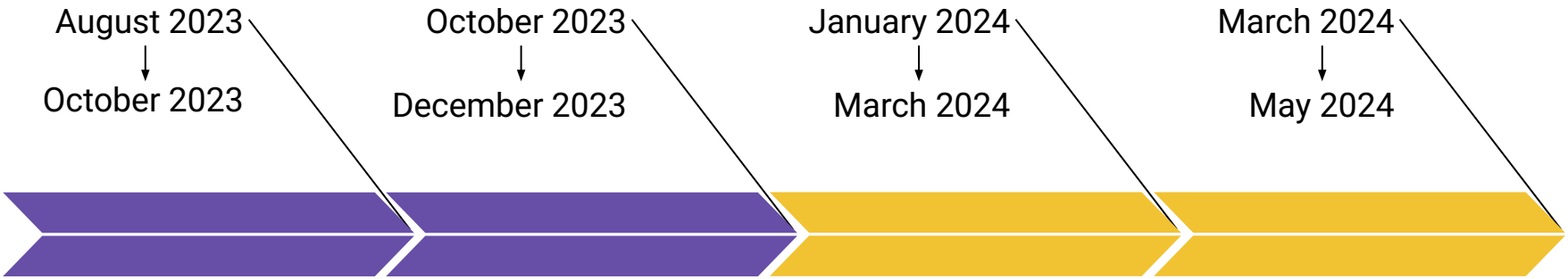


What are the parameters of the design?

- 2024 IGVC Rules (Link)
- Budget $5,000 + Fundraising
- August 2023 - May 2024



Timeline



Research Phase

Conduct research on necessary components, software, and algorithms

Individual Subsystem Development

Work on assigned component with the goal to have a standalone subsystem for the robot

Subsystem Integration

Combine all of the individual subsystems to create a functioning navigation system

Algorithm Development

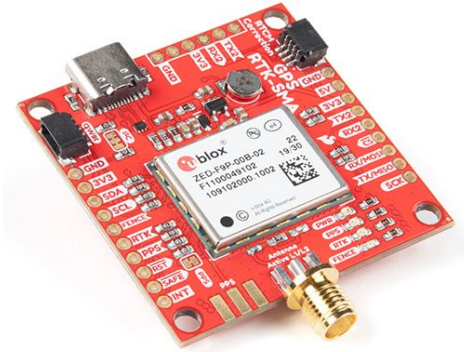
With the combined system, develop the means of which the robot will navigate

First Semester **Second Semester**

Research Component - GPS Receiver

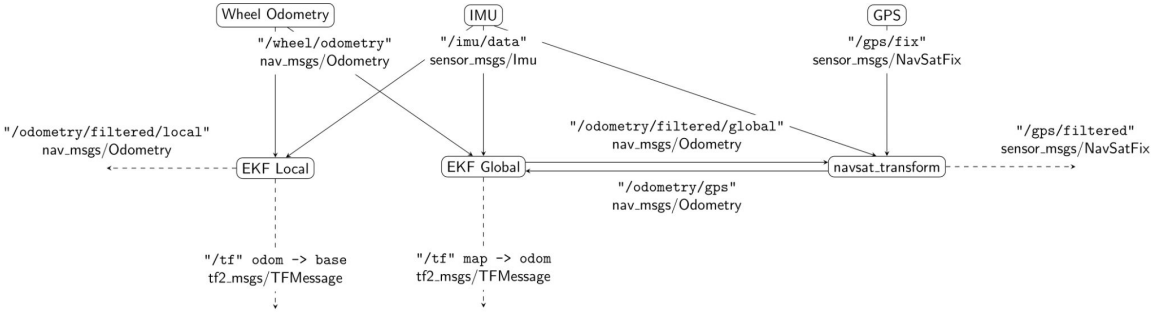
Purpose and Use

- Receiver for GPS Coordinates
- Necessary for Waypoint Navigation
- Fed into Extended Kalman Filter



Research

- Previous IGVC Teams
- Different Components
- Kalman Filter



GPS Research Results

Unit	Cost	Accuracy	Rank
SparkFun ZED-F9P Application Board	$274.95	1 cm	1 ✅
ublox C009-F9P Application Board	$301.79	1 cm	2
ublox C94-M8P Application Board	$399.99	2.5 m	3
EMLID Reach M2	$649	7 mm	4
Goouu Tech GT-U7 Module	$14.99	2.5 m	5

Software Research

Different Areas of Research

- Robot Operating System
- Open Source Code
- Python, C++, Arduino Language

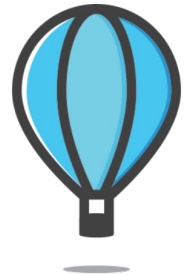
Robot Operating System (ROS)

- ROS vs. ROS2
- Different Releases
- How to Use and Operate

ROS2

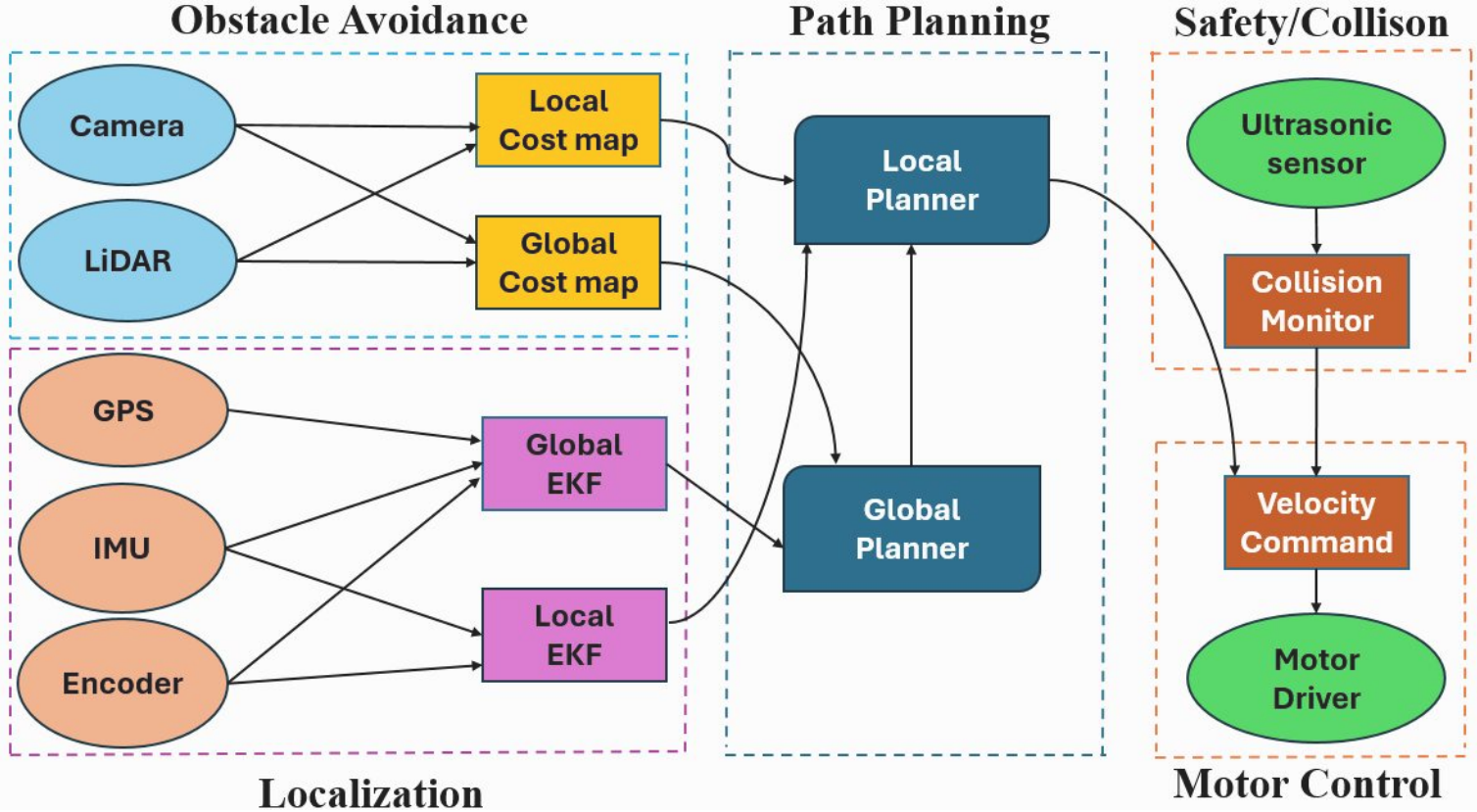






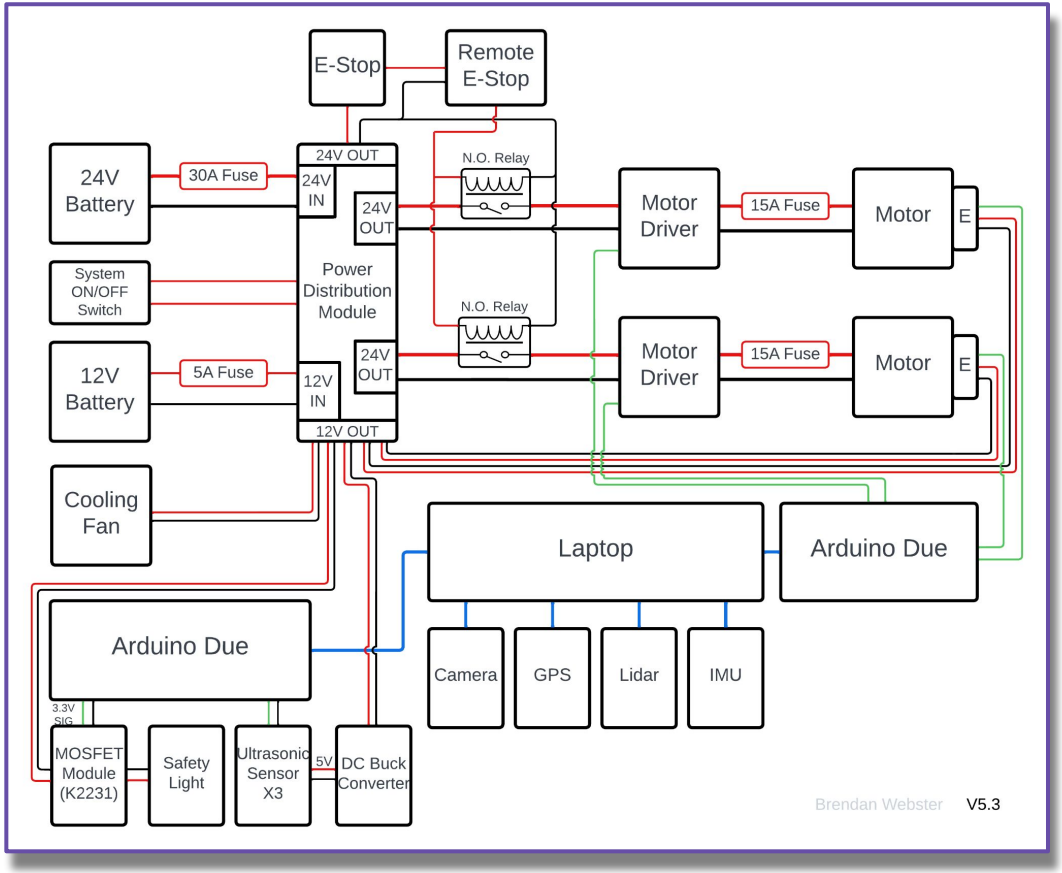
N A V 2

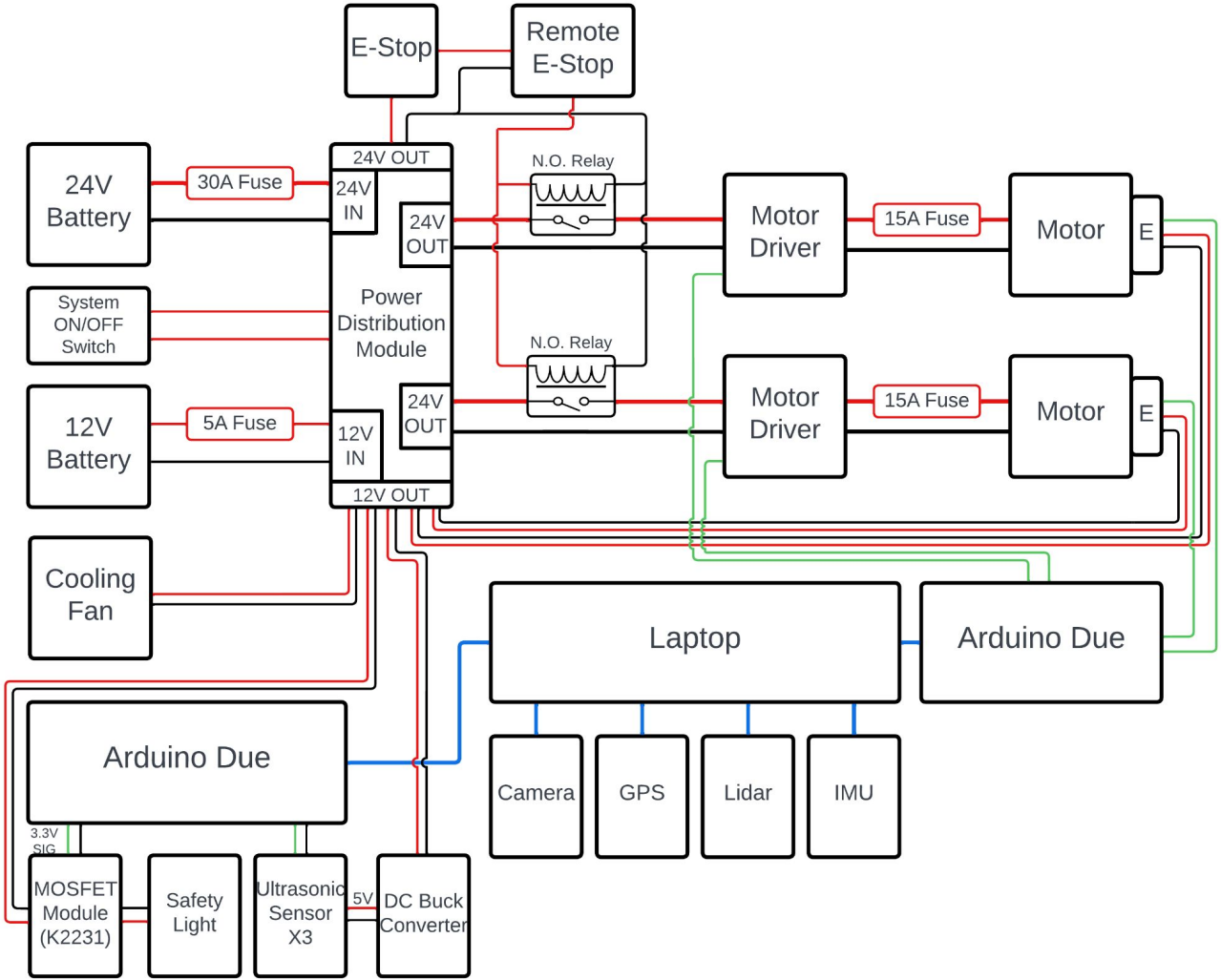
Final Navigation Prototype



Wiring Diagram

	Positive
	Negative, GND
	Data/Signal
	USB



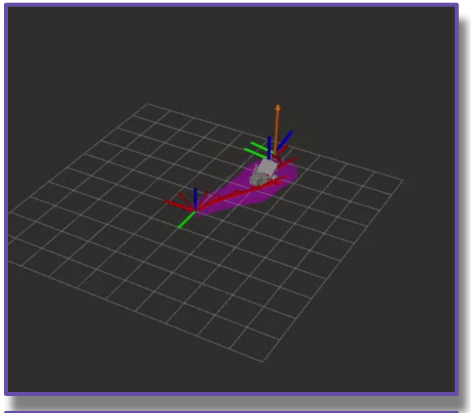

E-Stop
Remote E-Stop
24V Battery
30A Fuse
24V OUT
24V IN
24V OUT
N.O. Relay
Motor Driver
15A Fuse
Motor
E
System ON/OFF Switch
Power Distribution Module
N.O. Relay
12V Battery
5A Fuse
12V IN
24V OUT
Motor Driver
15A Fuse
Motor
E
12V OUT
Cooling Fan
Laptop
Arduino Due
Arduino Due
Camera
GPS
Lidar
IMU
3.3V
SIG
MOSFET Module (K2231)
Safety Light
Ultrasonic Sensor X3
5V
DC Buck Converter

Localization

Extended Kalman Filter (EKF)

- Fuses Data from Multiple Sources
- Local (IMU and Encoders)
- Global (IMU, Encoders, GPS)

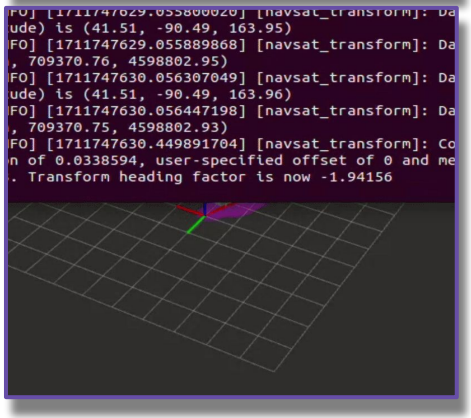
Local



Waypoint Navigation

- Uses Output of Global EKF
- Coordinates and Orientation
- Multiple Points

Global



Camera

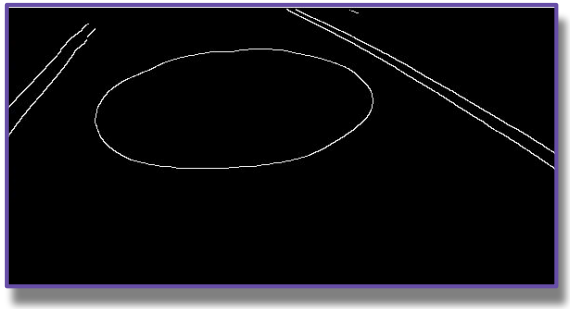
Purpose and Use

- Line and Pothole Detection
- Used for Obstacle Avoidance



How it is Used

- Raw Camera Data is converted to PointCloud
- White is Filtered from PointCloud Data
- Converted into LiDAR Compatible Data Type



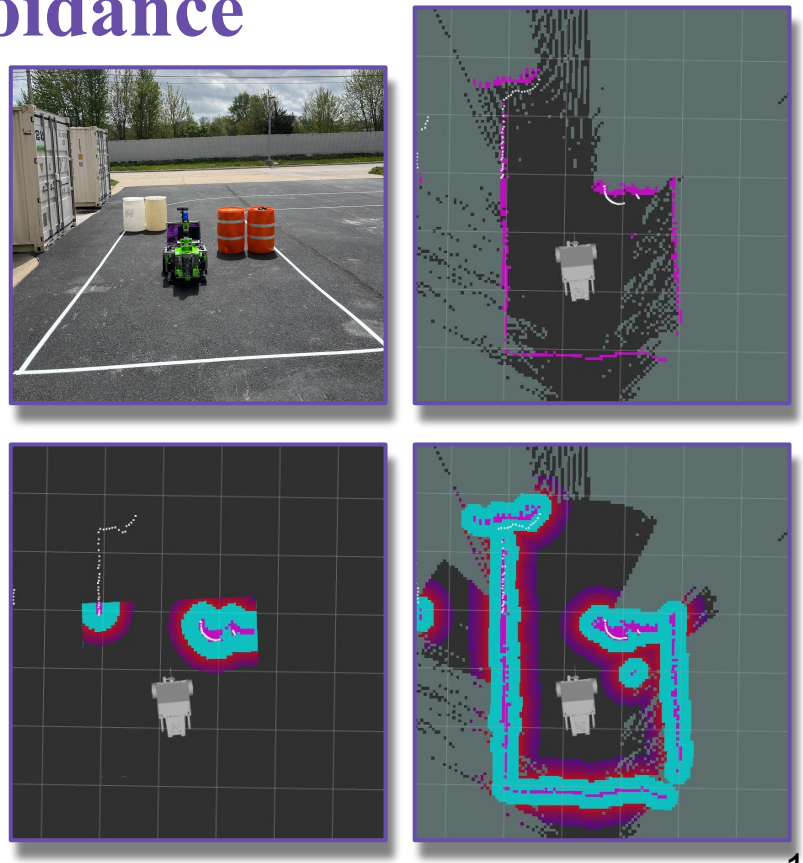
Obstacle Avoidance

Camera and LiDAR Integration

- Combine Sensors' Data
- Treat all Perceived Data as Obstacles
- All Obstacles saved using SLAM

Costmaps

- Add Cost around each Obstacle
- Local and Global Costmaps Created
- Used for Path Planning



Path Planning

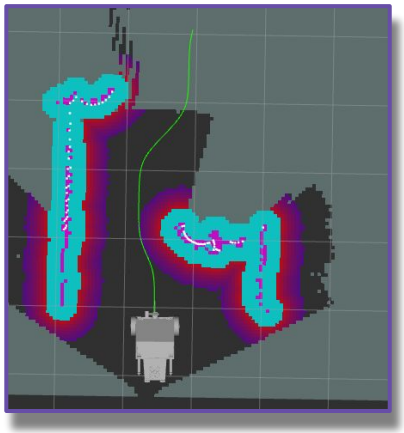
Global Planner

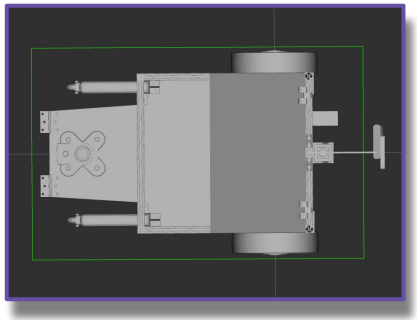
- Nav2 Smac Planner
- Long-term Waypoint Navigation
- Saves all Cost Data for Map

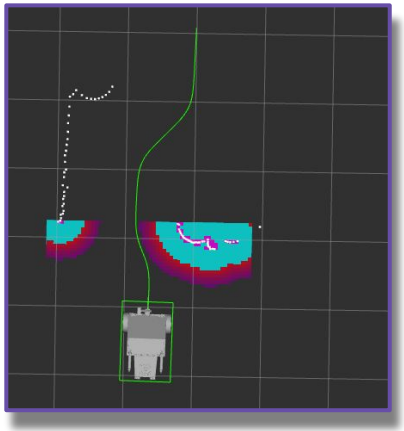
Local Planner

- DWB Controller
- Short-term Obstacle Avoidance
- Provides Velocity Commands









Final Vehicle Design





Final Product



Total Expenses:

Purchase Total:

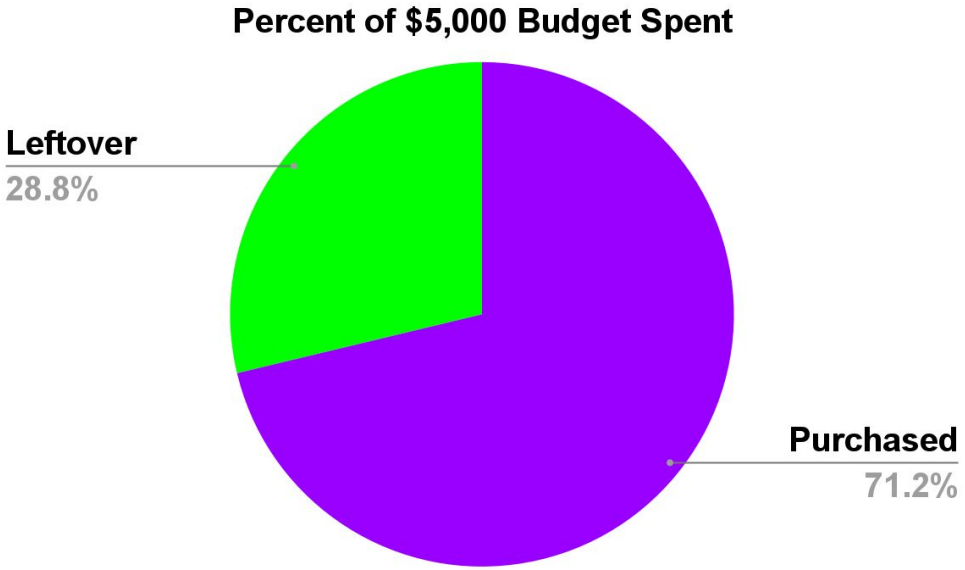
$3,562.48

No Purchase Total:

$3,143.71

Gross Total:

$6,706.19



Questions?

